Università
di Genova

High Performance Computing

Report CUDA

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Contents

1 Architectures

For this project, I used two workstations: my personal laptop and the Google Colab environment

1.1 HP Envy 17-ae103nl (Laptop)

CPU

- **Model**: Intel Core i7-6500U @ 2.50 GHz
- **Architecture**: x86_64
- **Physical cores**: 2
- **Logical threads**: 4 (Hyper-Threading)

GPU

- **Model**: NVIDIA GeForce GTX 950M
- **CUDA Compute Capability**: 5.0
- **Driver version**: 572.16
- **CUDA Version**: 12.8 (*CUDA Toolkit 12.x*)

Software and Setup:

- Visual Studio Community 2022 - Workload: Desktop development with C++
- CUDA Toolkit installed manually to enable GPU programming
- WSL2
- NVIDIA drivers (version 572.16) installed for WSL2 from NVIDIA official website

1.2 Google Colab

CPU

- **Model**: Intel(R) Xeon(R) CPU @ 2.20GHz
- **Architecture**: x86_64
- **Physical cores**: 1
- **Logical threads**: 2 (Hyper-Threading)
- **Socket(s)**: 1

GPU

- **Model**: NVIDIA Tesla T4
- **CUDA Compute Capability**: 7.5
- **Driver version**: 550.54.15
- **CUDA Version**: 12.8 (*CUDA Toolkit 12.x*)

2 Hotspot Analysis

Heat conduction (or heat diffusion) is the process by which thermal energy spreads within a material due to temperature differences. In two dimensions, this phenomenon can be represented as heat spreading across a plate or surface over time. Numerical simulations of heat conduction are widely used in engineering, physics, and high-performance computing to predict temperature distribution and analyze thermal effects.

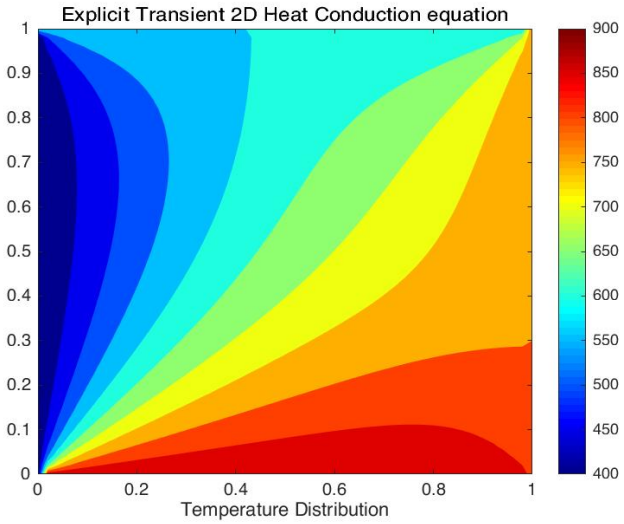


Figure 1: Illustration of 2D heat conduction simulation.

The formula to represent 2D heat conduction is given as follows:

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \tag{1}$$

where:

- T is temperature;
- t is time;
- α is the thermal diffusivity;
- (x, y) are points in a grid.

To solve this problem numerically, one possible finite difference approximation is:

$$\frac{\Delta T}{\Delta t} = \frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t} = \alpha \left(\frac{T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n}}{(\Delta x)^2} + \frac{T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n}}{(\Delta y)^2} \right) \tag{2}$$

where:

- Δt is the timestep;
- i, j are indices in the grid;
- n indicates the current timestep, so $T_{i,j}^{n}$ is the temperature at point (i, j) at time $t = n\Delta t$, and $T_{i,j}^{n+1}$ is the temperature at the next timestep $t = (n+1)\Delta t$.

2.1 Code Description

This program simulates the diffusion of heat on a two-dimensional plate using the finite difference method. It includes two main implementations:

1. **Reference CPU version** - `step_kernel_ref()` - serial implementation
2. **Modified version** - `step_kernel_mod()` - intended for CUDA acceleration

Both implementations are designed to produce the same results.

Data Layout

The simulation uses a grid of size `ni × nj` (columns × rows).
Temperature values are stored in a **1D array**. A macro is used to convert 2D indices to linear index:

```
#define I2D(num, c, r) ((r)*(num)+(c))
```

Initialization

In the beginning, the boundary points of the grid are assigned with initial *random* temperatures, while the inner points update their temperature iteratively according to the formula above.

```
  for (int i = 0; i < ni * nj; ++i) {
      t1_ref[i] = t2_ref[i] =
      t1[i]     = t2[i]     =
          (float)rand() / (float)(RAND_MAX / 100.0f);
  }
```

Realistic System Inizialization

In addition to the synthetic configurations used, I also decided to simulate a more realistic scenario in order to visually inspect the physical behavior of the system. Specifically, I initialized the domain by placing a high–temperature square region at the center of the grid, representing a localized heat source. All remaining points were initialized with a low random temperature, mimicking environmental thermal noise. The initialization code is reported below:

```
// Hot square parameters
int hot_size = 40;     // side length of the hot square
int hot_temp = 80;     // temperature of the hot region
int hot_i0 = ni/2 - hot_size/2;     // center coordinate in i
int hot_j0 = nj/2 - hot_size/2;     // center coordinate in j

for(int j = 0; j < nj; j++){
  for(int i = 0; i < ni; i++){
    int idx = I2D(ni, i, j);

    if(i >= hot_i0 && i < hot_i0 + hot_size &&
```

```
12          j >= hot_j0 && j < hot_j0 + hot_size)
13      {
14        // heat source
15        temp1[idx] = temp2[idx] = (float)hot_temp;
16      }
17      else
18      {
19        // random background
20        temp1[idx] = temp2[idx] = (float)rand()/RAND_MAX*30.0f;
21      }
22    }
23  }
```

The source code can be found in `src/image.cu` and the script used for the visualization was done with Matlab (more details in: `src/realistic_simulation.m`)

The visual outcome of this configuration is discussed in Subsection 4.1.

Execution

The core of the simulation is inside the `step_kernel()` function, which implements the approximation described in formula (2), shown below.

```
1  void step_kernel_mod(int ni, int nj, float fact, float* temp_in,
       float* temp_out)
2  {
3    int i00, im10, ip10, i0m1, i0p1;
4    float d2tdx2, d2tdy2;
5
6    // loop over all points in domain (except boundary)
7    for ( int j=1; j < nj-1; j++ ) {
8      for ( int i=1; i < ni-1; i++ ) {
9        // find indices into linear memory
10       // for central point and neighbours
11       i00 = I2D(ni, i, j);
12       im10 = I2D(ni, i-1, j);
13       ip10 = I2D(ni, i+1, j);
14       i0m1 = I2D(ni, i, j-1);
15       i0p1 = I2D(ni, i, j+1);
16
17       // evaluate derivatives
18       d2tdx2 = temp_in[im10]-2*temp_in[i00]+temp_in[ip10];
19       d2tdy2 = temp_in[i0m1]-2*temp_in[i00]+temp_in[i0p1];
20
21       // update temperatures
22       temp_out[i00] = temp_in[i00]+fact*(d2tdx2 + d2tdy2);
23     }
24   }
25 }
```

Note: The formula corresponds (2) to lines 18–23 in the above code.

Verification

After all timesteps, `temp1_ref` holds the CPU reference results, while `temp1` holds the modified version results. If the error exceed a threeshold (e.g. 0.0005), the simulation is considered correct.

```
1    float maxError = 0;
2
3    for( int i = 0; i < ni*nj; ++i ) {
4      if (abs(temp1[i]-temp1_ref[i]) > maxError) { maxError = abs(
         temp1[i]-temp1_ref[i]); }
5    }
6
7    // Check and see if our maxError is greater than an error bound
8    if (maxError > 0.0005f)
9      printf("The error is NOT within acceptable bounds.");
10   else
11     printf("The error is within acceptable bounds");
```

2.2 Project Structure

The `homework2` project is organized into four main directories:

- **build**: contains the compiled executables.
- **reports**: stores the reports generated with the `-qopt-report` flag.
- **src**: includes the source code files.
- **metrics**: scripts that run the executables in `build` and save performance metrics.

2.3 Profiling

For performance profiling, I chose to use the university workstation, using Intel's profiling tools: **VTune** and **Advisor**. The code was compiled using the Intel Compiler (**ICX**) with the following flags: `-O3 -g -xHost` to enable high optimization, generate debug information, and target the host CPU architecture.

Simulation Parameters

The analysis was conducted considering data size resulting in a sequential execution time of ~ 30 seconds, by setting:

- $nstep = 200$
- $ni = 20'000$
- $nj = 20'000$
- $size = ni \times nj = 400'000'000$

Compiler Reports

In addition to runtime profiling, I also examined the compiler optimization reports generated with the flag `-qopt-report`. These reports provide insights on which loops and functions the compiler was able to optimize or vectorize.

Advisor Roofline Analysis

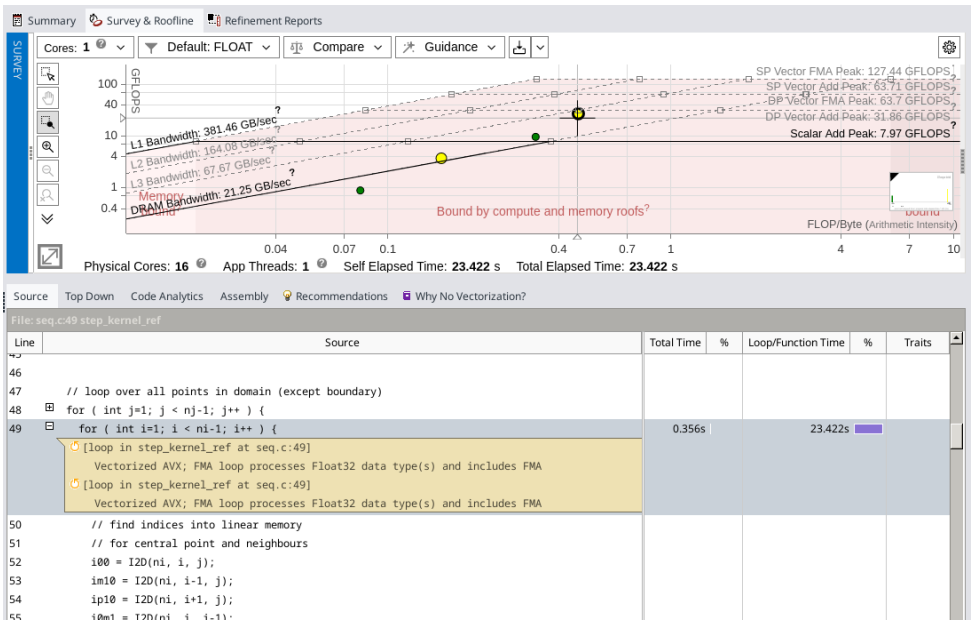


Figure 2: Roofline analysis of the sequential baseline code.

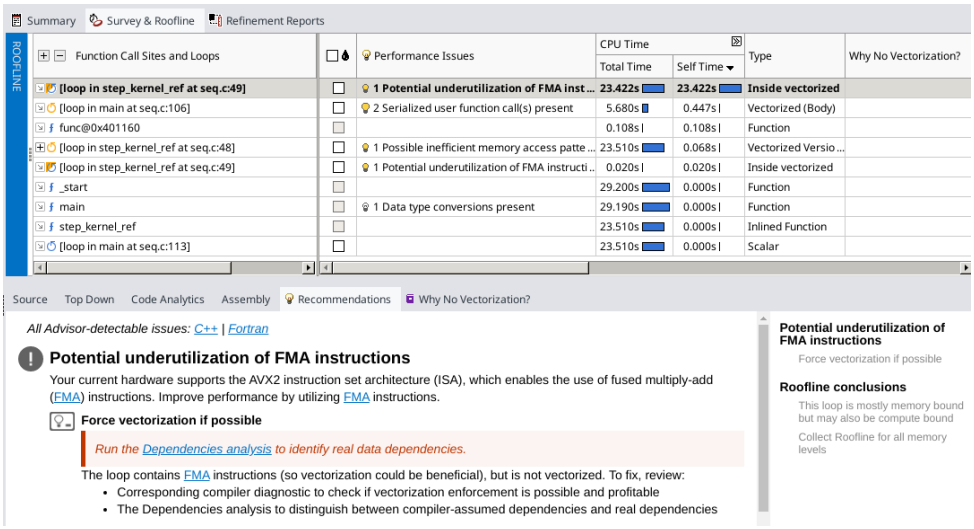


Figure 3: Advisor Survey.

TODO: aggiungere una breve descrizione di queste due immagini

3 Parallelization

3.1 Sequential section

Ho deciso inizialmente di eseguire i test sul mio PC personale, piuttosto che sulla workstation dell'università o su Google Colab, per evitare limitazioni dovute a sessioni temporanee e risorse limitate. In questo modo ho avuto maggiore controllo sul processo di compilazione e sull'esecuzione dei test.

Questa parte del report fa riferimento solo all'analisi eseguita tramite il mio PC personale. I parametri della simulazione utilizzati sono:

- $nstep = 200$
- $ni = 15000$
- $nj = 15000$
- $size = ni \times nj = 900\,000\,000$

Ho compilato il codice sequenziale presente in `src/cuda.c` using the following options:

```
gcc -O3 src/cuda.c -o build/cuda
```

Il tempo di esecuzione migliore ottenuto ($37s$) costituirà il riferimento sequenziale per il calcolo dello speedup nella versione GPU.

3.2 CUDA

3.2.1 Code

La parallelizzazione è stata realizzata utilizzando CUDA per eseguire il calcolo dei passi temporali su GPU. Il kernel principale è `step_kernel_mod`, che replica il comportamento della funzione sequenziale `step_kernel_ref`.

Il codice si può trovare in `src/cuda.cu`

Struttura del kernel:

- Ogni thread gestisce un punto della griglia 2D.
- Gli indici globali i e j vengono calcolati a partire dai `threadIdx` e `blockIdx`.

Gestione della memoria:

- Allocazione della memoria host e device.
- Copia dei dati iniziali dal host alla device.
- Copia dei risultati dalla device al host al termine della simulazione.

Controllo errori:

- Dopo ogni chiamata al kernel viene eseguito `cudaGetLastError()`.
- Viene calcolato l'errore massimo rispetto al riferimento CPU per verificare la correttezza.

```
__global__ void step_kernel_mod(int ni, int nj, float fact, float
    * temp_in, float* temp_out)
{
  int i00, im10, ip10, i0m1, i0p1;
  float d2tdx2, d2tdy2;

  int i = threadIdx.x + blockIdx.x * blockDim.x;
  int j = threadIdx.y + blockIdx.y * blockDim.y;

  if(i > 0 && i < ni-1 && j > 0 && j < nj-1){
    // find indices into linear memory
    // for central point and neighbours
    i00 = I2D(ni, i, j);
    im10 = I2D(ni, i-1, j);
    ip10 = I2D(ni, i+1, j);
    i0m1 = I2D(ni, i, j-1);
    i0p1 = I2D(ni, i, j+1);

    // evaluate derivatives
    d2tdx2 = temp_in[im10]-2*temp_in[i00]+temp_in[ip10];
    d2tdy2 = temp_in[i0m1]-2*temp_in[i00]+temp_in[i0p1];

    // update temperatures
    temp_out[i00] = temp_in[i00]+fact*(d2tdx2 + d2tdy2);
  }
}
```

3.2.2 Compilation

La compilazione CUDA è stata effettuata con il comando:

```
nvcc -O3 -arch=sm_50 cuda.cu -o cuda.exe -DDEBUG=0
```

Significato dei flag:

- `-O3`: ottimizzazione massima.
- `-arch=sm_50`: architettura target della GPU.
- `-DDEBUG=0`: disabilita il debug per migliorare le performance.

3.2.3 Execution

Per il kernel CUDA, la scelta di `threadsPerBlock` e `numBlocks` è stata la seguente:

```
    dim3 threadsPerBlock(16, 16);
    dim3 numBlocks((ni + 15) / 16, (nj + 15) / 16);

    // ---- CUDA events start ----
    cudaEvent_t start, stop;
    cudaEventCreate(&start);
    cudaEventCreate(&stop);

    // Execute the modified version using same data
    printf("Execute GPU kernel...\n");
    cudaEventRecord(start, 0); // start timing

    for (istep = 0; istep < nstep; istep++) {
        step_kernel_mod <<<numBlocks, threadsPerBlock>>> (ni, nj,
            tfac, d_temp1, d_temp2);
        cudaDeviceSynchronize();

        cudaError_t err = cudaGetLastError();
        if (err != cudaSuccess)
            printf("CUDA Error: %s\n", cudaGetErrorString(err));

        float* d_tmp = d_temp1;
        d_temp1 = d_temp2;
        d_temp2 = d_tmp;
    }

    cudaEventRecord(stop, 0); // stop timing
    cudaEventSynchronize(stop);
```

Ogni blocco di 16x16 thread calcola i punti della griglia, mentre i blocchi sono organizzati per coprire l'intera matrice.
La temporizzazione è stata eseguita tramite `cudaEvent_t` per misurare il tempo del kernel GPU.

3.2.4 Results

Profiling (command: `nvprof build/cuda`)

950M:

```
Execute the CPU-only reference version...
- Elapsed time: 0.188 s
Execute GPU kernel...
- Elapsed time: 140.515 ms
Max Error = 0.000008
The Max Error of 0.00001 is within ACCEPTABLE bounds.
- Total elapsed time: 2.460 s
==768== Profiling application: build/cuda
==768== Profiling result:
Type  Time(%)      Time     Calls       Avg       Min       Max  Name
GPU:   72.95%  202.16ms         2  101.08ms  2.7665ms  199.40ms  [CUDA memcpy HtoD]
       24.93%  69.101ms       200  345.50us  342.59us  355.94us  step_kernel_mod(int, int, float, float*, float*)
        2.12%  5.8774ms         2  2.9387ms  2.5962ms  3.2812ms  [CUDA memcpy DtoH]
```

```
API:   90.99%  1.69655s         2  848.27ms  699.40us  1.69585s  cudaMalloc
        7.10%  132.39ms       200  661.97us  423.90us  7.6760ms  cudaDeviceSynchronize
        0.96%  17.932ms         4  4.4831ms  3.1292ms  7.7131ms  cudaMemcpy
        0.45%  8.4117ms         2  4.2059ms  2.4000us  8.4093ms  cudaEventCreate
        0.43%  8.0093ms       200  40.046us  9.9000us  1.0583ms  cudaLaunchKernel
        0.04%  796.50us         2  398.25us  302.30us  494.20us  cudaFree
        0.01%  213.70us       200  1.0680us     200ns  31.300us  cudaGetLastError
        0.00%  90.700us         1  90.700us  90.700us  90.700us  cudaEventSynchronize
        0.00%  81.700us       101     808ns     200ns  31.300us  cuDeviceGetAttribute
        0.00%  30.100us         2  15.050us  10.700us  19.400us  cudaEventRecord
        0.00%  5.7000us         2  2.8500us  1.3000us  4.4000us  cudaEventDestroy
        0.00%  4.3000us         1  4.3000us  4.3000us  4.3000us  cuDeviceGetPCIBusId
        0.00%  3.9000us         1  3.9000us  3.9000us  3.9000us  cudaEventElapsedTime
        0.00%  2.3000us         3     766ns     400ns  1.4000us  cuDeviceGetCount
        0.00%  1.6000us         2     800ns     300ns  1.3000us  cuDeviceGet
        0.00%  1.5000us         1  1.5000us  1.5000us  1.5000us  cuDeviceGetName
        0.00%     800ns         1     800ns     800ns     800ns  cuDeviceTotalMem
        0.00%     400ns         1     400ns     400ns     400ns  cuDeviceGetUuid
```

T4:

```
Execute the CPU-only reference version...
- Elapsed time: 36.761 s
Execute GPU kernel...
- Elapsed time: 1741.090 ms
Max Error = 0.000015
The Max Error of 0.00002 is within ACCEPTABLE bounds.
- Total elapsed time: 46.074 s
==5152== Profiling application: /content/drive/MyDrive/Colab Notebooks/HPC-CUDA/build/cuda
==5152== Profiling result:
Type  Time(%)      Time     Calls       Avg       Min       Max  Name
GPU:   67.30%  1.73103s       200  8.6551ms  8.0181ms  11.542ms  step_kernel_mod(int, int, float, float*, float*)
       16.79%  431.89ms         2  215.95ms  203.45ms  228.45ms  [CUDA memcpy HtoD]
       15.91%  409.13ms         2  204.56ms  201.12ms  208.01ms  [CUDA memcpy DtoH]
API:   62.34%  1.73679s       200  8.6840ms  8.0240ms  11.548ms  cudaDeviceSynchronize
       30.23%  842.24ms         4  210.56ms  201.46ms  228.69ms  cudaMemcpy
        7.16%  199.38ms         2  99.688ms  144.08us  199.23ms  cudaMalloc
        0.13%  3.7350ms       200  18.674us  5.4010us  429.51us  cudaLaunchKernel
        0.13%  3.6890ms         2  1.8445ms  870.86us  2.8181ms  cudaFree
        0.01%  168.47us       114  1.4770us     185ns  65.476us  cuDeviceGetAttribute
        0.00%  67.106us       200     335ns     105ns     674ns  cudaGetLastError
        0.00%  33.409us         2  16.704us  10.078us  23.331us  cudaEventRecord
        0.00%  27.898us         2  13.949us  1.1280us  26.770us  cudaEventCreate
        0.00%  12.296us         1  12.296us  12.296us  12.296us  cuDeviceGetName
        0.00%  6.8260us         1  6.8260us  6.8260us  6.8260us  cuDeviceGetPCIBusId
        0.00%  4.4120us         1  4.4120us  4.4120us  4.4120us  cudaEventSynchronize
        0.00%  3.6190us         1  3.6190us  3.6190us  3.6190us  cudaEventElapsedTime
        0.00%  3.3080us         2  1.6540us  1.1110us  2.1970us  cudaEventDestroy
        0.00%  1.6790us         3     559ns     233ns  1.2100us  cuDeviceGetCount
        0.00%     716ns         2     358ns     179ns     537ns  cuDeviceGet
        0.00%     529ns         1     529ns     529ns     529ns  cuDeviceTotalMem
        0.00%     372ns         1     372ns     372ns     372ns  cuModuleGetLoadingMode
        0.00%     302ns         1     302ns     302ns     302ns  cuDeviceGetUuid
```

I risultati della simulazione GPU sono stati confrontati con la versione CPU e lo speedup è calcolato come:

$$\text{speedup} = \frac{T_{\text{CPU}}}{T_{\text{GPU}}}$$

GPU	$\mathbf{T_{CPU}}$ (ms)	$\mathbf{T_{GPU}}$ (ms)	**Speedup**
950M	34967	15086	x2.32
T4	37178	1645	x22.60

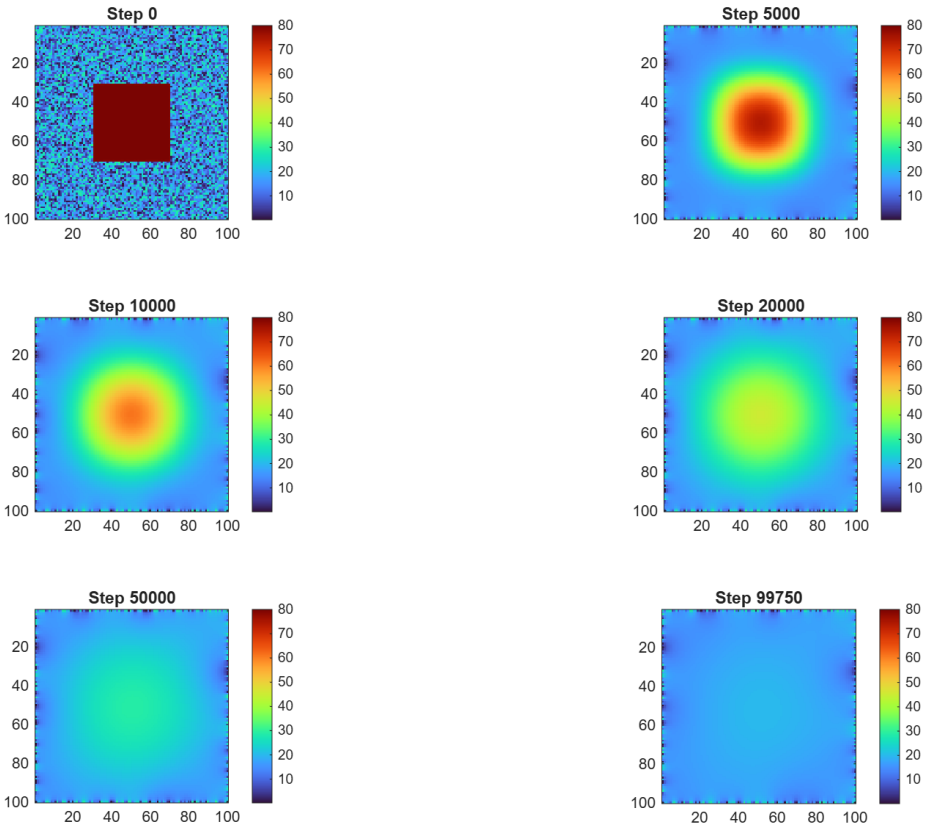
Table 1: Confronto tra tempi CPU e GPU e speedup

4 Conclusions

4.1 Realistic System Simulation

A seguito delle analisi descritte nel paragrafo precedente, ho voluto integrare il progetto con una simulazione grafica reale, per dimostrare che la simulazione è corretta.
I parametri della simulazione reale utilizzati sono:

- $nstep = 100'000$
- $step = 250$
- $ni = 100$
- $nj = 100$
- $tfac = 0.01f$
- $size = ni \times nj = 10'000$



4.2 Considerations

La parallelizzazione tramite GPU ha permesso di ottenere uno speedup significativo, dimostrando l'efficacia dell'approccio rispetto alla CPU.

L'errore massimo tra GPU e CPU è stato calcolato per verificare la correttezza dei risultati. Nella simulazione corrente, il massimo errore osservato era entro i limiti accettabili (0.0005).

I risultati confermano che la parallelizzazione con CUDA riduce significativamente i tempi di esecuzione rispetto alla versione sequenziale, mantenendo la precisione numerica.